01/10/2026

PRACTICAL - 9

Aim:- Implementation of SUBNETTING in CISCO PACKET TRACER Simulator.

- Creating Network Topology

Select 'New' Button in top left corner & select "Network" & "Generic". This will create a blank network topology that we can use to add devices.

- Adding the devices

To add a device, select device from bottom left corner & drag it onto network topology, then connect devices by dragging cable from one device's port to another devi's port

- Subnetting

To subnet the address of 192.168.1.0/24 to provide enough space for at least 5 addresses for end devices, the switch, the router, we can use a 127 subnet mask.

- Configuring the devices

Right click on the router & Select "CLI". This will open the command-line Interface for the router. Enter the commands!

enable
configure terminal
Interface FastEthernet0/0
ip address {IP address} {subnet mask}
no shutdown
exit

Interface FastEthernet0/1
ip address {IP address} {subnet mask}
no shutdown
exit

Configuring switch:

Righ-click on the switch & select "CLI"
enable
Configure terminal
Interface FastEthernet0/1
Switchport mode access
exit
Interfaces FastEthernet0/2
switchport mode access
exit

Configuring the PC's:

Right click on PC & select "Config":

Enter IP address, subnet mask, default default gateway & DNS server

information.

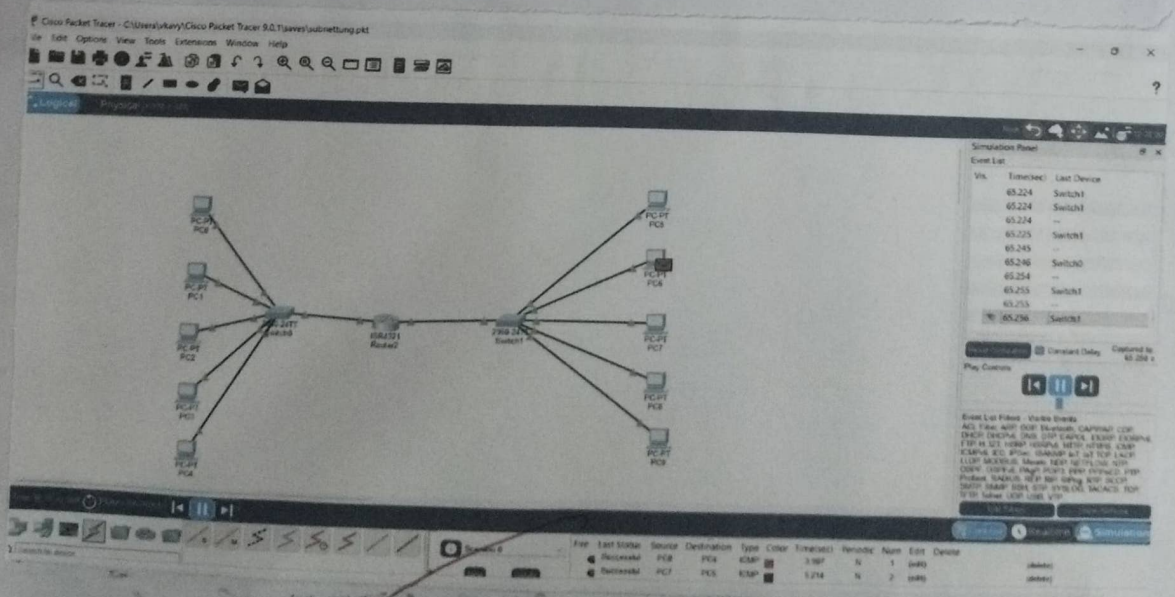
Configuring GigabitEthernet Interface:

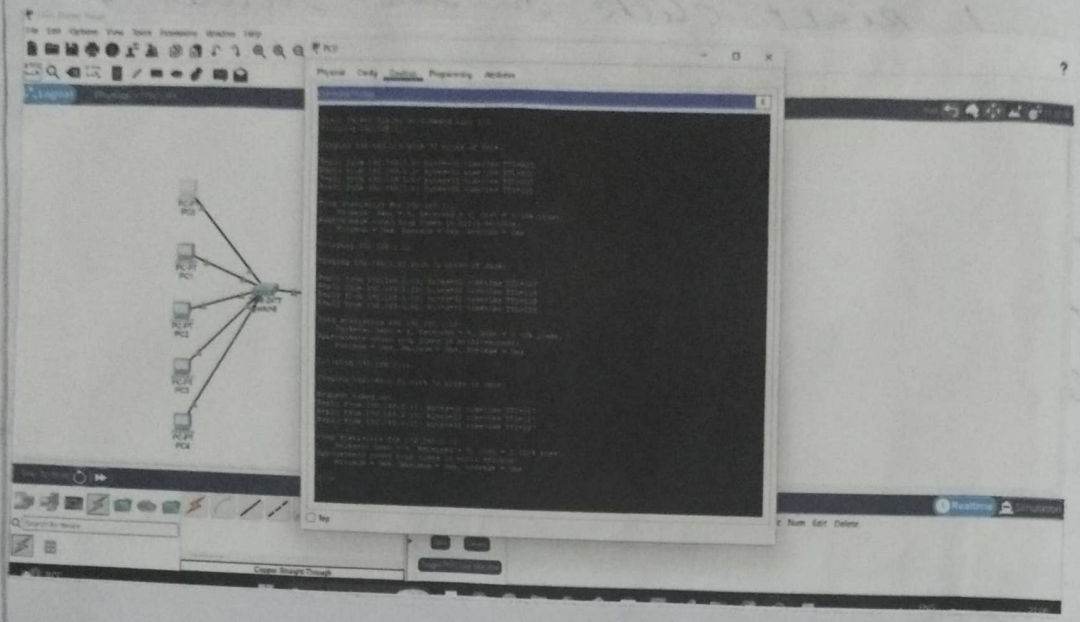
1. Right-click on the router & Select "CLI".
2. Enter the following

Commands: enable
configure terminal
interface GigabitEthernet 0/0
ip address {IP address} {subnet mask}
no shutdown
exit

Testing the network:

Open a command prompt on each PC & try to ping the other PC.
If ping is successful, then network is functioning properly.





✓ 8/16

Result: Thus, SUBNETTING in Simulator CISCO Packet Tracer was Implemented successfully.